AI for Lexical & Syntax Assistance

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WEEK 2 REPORT: Literature Survey & Gap Identification

1. **Introduction** A comprehensive literature survey was conducted to understand the current state of "Automated Program Repair" (APR). The study focused on identifying existing tools, their limitations, and the specific technologies required to build an AI-based syntax assistant.

2. Literature Review (Existing Systems)

   - DeepFix (Gupta et al., 2017):

     - *Description:* A deep learning-based tool developed by researchers to fix common C language programming errors. It treats code correction as a machine translation problem (translating "bad code" to "good code").

     - *Limitation:* It requires massive computational power and is not easily accessible as a lightweight student tool.

   - Standard Compiler Diagnostics (GCC/Clang):

     - *Description:* Modern compilers use sophisticated parsing algorithms (like LL or LR parsers) to find errors.

     - *Limitation:* They rely on rigid rules. If a user makes a mistake that doesn't fit the rule, the compiler often gives a "syntax error" at the wrong line number without a fix suggestion.

- TRACER (Repairing Compilation Errors):
    - *Description:* A tool that uses comparison-based methods to find errors.
    - *Limitation:* It struggles with multiple errors on a single line.

Feature	**Standard Compilers (GCC)**	**DeepFix (Research Tool)**	**Proposed System (Our Project)**
Error Detection	Rule-based (Rigid)	AI-based (Flexible)	AI-based (Flexible)
Auto-Correction	No	Yes	Yes
Target Audience	Experts	Researchers	Students/Beginners
Complexity	Low (Built-in)	Very High	Medium (Lightweight)

- **Gap Identification** Based on the survey, the following gaps were identified:
    - **Accessibility:** Most AI repair tools are research prototypes and not integrated into simple student workflows.
    - **Explainability:** Existing tools fix the code but do not explain *why* the fix was made.
    - **Scope:** Many tools focus on complex logic errors rather than the simple syntax errors that frustrate beginners the most.

- **Technology Selection** To address these gaps, the following technology stack has been selected for this project:

  - **Programming Language:** Python (for its extensive NLP libraries).

  - **Machine Learning Model: LSTM (Long Short-Term Memory)** networks, as they are effective at remembering long sequences of code tokens.

  - **Frameworks:** TensorFlow or PyTorch for model development.

  - **Dataset:** Prutor dataset or GitHub repositories for C/Python source code.